

Regular Article

Increased beta power in the bereaved families of the Sewol ferry disaster: A paradoxical compensatory phenomenon? A two-channel electroencephalography study

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Aim: The Sewol ferry capsizing accident on South Korea's southern coast resulted in the death of 304 people, and serious bereavement problems for their families. Electroencephalography (EEG) beta frequency is associated with psychiatric symptoms, such as insomnia. The aim of this study was to investigate the relation between frontal beta power, psychological symptoms, and insomnia in the bereaved families.

Methods: Eighty-four family members of the Sewol ferry victims (32 men and 52 women) were recruited and their EEG was compared with that of 25 (13 men and 12 women) healthy controls. A two-channel EEG device was used to measure cortical activity in the frontal lobe. Symptom severity of insomnia, post-traumatic stress disorder, complicated grief, and anxiety were evaluated.

Results: The bereaved families showed a higher frontal beta power than healthy controls.

Subgroup analysis showed that frontal beta power was lower in the individuals with severe insomnia than in those with normal sleep. There was a significant inverse correlation between frontal beta power and insomnia symptom in the bereaved families.

Conclusion: This study suggests that increased beta power, reflecting the psychopathology in the bereaved families of the Sewol ferry disaster, may be a compensatory mechanism that follows complex trauma. Frontal beta power could be a potential marker indicating the severity of sleep disturbances. Our results suggest that sleep disturbance is an important symptom in family members of the Sewol ferry disaster's victims, which may be screened by EEG beta power.

Key words: bereavement, beta power, electroencephalography, insomnia, Sewol ferry disaster.

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ON 16 APRIL 2014, the Sewol ferry sank on the southern coast of South Korea, resulting in the death of 304 people; 250 victims were 16–17-year-old high-school students who were on a school trip. The Sewol ferry disaster caused sudden severe complicated bereavement in families of the victims, who witnessed the accident through news broadcasts. The South Korean government called the disaster a

national tragedy, and the society still grieves for those lost.

Electroencephalography (EEG) is a non-invasive method that records the electrical activity of the brain from the scalp. A two-channel EEG device has been used to explore spectral information with high accuracy in the human brain.^{1–4} It has been reported that EEG activity change is associated with psychiatric disorders.^{5–11} Lee *et al.* reported that the functional connectivity of beta frequency was decreased in patients with post-traumatic stress disorder (PTSD) compared to healthy controls, and beta connectivity was correlated with PTSD symptoms.¹² Changes in the beta band have been related to panic disorder as well.¹³ A review by Carvalho *et al.*¹³ reported that there is a propensity of elevated beta power in patients with panic disorder. Its results indicate that enhanced beta band is associated with a state of hyperarousal to the lack of cognitive flexibility and control.

Meanwhile, a study has demonstrated the modulation effect of EEG beta frequency on primary insomnia.¹⁴ It found that beta 1 and beta 2 activities increased in veterans with PTSD when compared to healthy subjects¹⁵ and concluded that increased beta activity contributes to cortical excitability, sleep problems, and attention disturbances. In the study by van Wyk *et al.*,¹⁶ hyperarousal symptoms predicted impaired sleep. The beta power was similarly increased during sleep in adolescents with insomnia, thereby indicating a level of cortical hyperarousal in these patients.¹⁷ However, neurophysiological alterations of beta activity in sleep disturbances are poorly understood in clinical settings. *In vivo*, the role of the neocortical area involved in beta oscillation during the wake–sleep cycle has been examined only in monkeys.¹⁸ Those authors found that beta frequency played a role as a highly coherent modulator by actively firing inhibitory neurons compared to excitatory neurons.

Hence, the goals of the present study were as follows: (i) to compare the frontal beta power between bereaved families (BF) and healthy controls (HC); and (ii) to investigate whether the frontal beta power was associated with psychological symptoms, such as sleep disturbances, in the families of the Sewol ferry victims. We hypothesized that the frontal beta power would be higher in the BF than in the HC. Second, we hypothesized that the frontal beta power of the BF would be correlated with insomnia.

METHODS

Eighty-four members of BF of victims of the Sewol ferry disaster (32 men and 52 women) were recruited through the affiliation with the Ansan Trauma Mental Health Center. BF without psychiatric past history were included in this study. The mean age was 44.95 ± 8.60 years (range, 19–60 years). Twenty-five HC without a current or lifetime history of psychiatric and neurological disorders were recruited (13 men and 12 women) through the community newspaper. The mean age was 43.44 ± 4.94 years (range, 38–56 years). Participants were assessed using an assessment package of self-report psychological rating scales. The descriptive statistics of the participants are summarized in Table 1. The duration of formal education (in years) was higher for the HC than for the BF (BF/HC: 12.52/15.36; $t = -5.81$, $P = 0.001$). The BF showed higher insomnia scores (BF/HC: 13.11/6.20; $t = 4.79$, $P = 0.001$) and general anxiety disorder (GAD) symptom scores (BF/HC: 10.63/2.28; $t = 8.56$, $P = 0.001$) than the HC. In order to determine the covariance, non-parametric (Spearman's rho) correlations were analyzed among EEG beta power, education, and anxiety symptoms in all subjects. In the BF, subgroups were classified into the low and high groups according to cut-off values of each of the psychological symptom scores. EEG power was compared between the low and high groups using *t*-tests. The BF were reanalyzed into four subgroups according to insomnia severity scores: normal sleep, 0–7; mild insomnia, 8–12; moderate insomnia, 13–17; and severe insomnia, 18–28. The study was approved by the Institutional Review Board of Seoul Saint Mary's Hospital, College of Medicine, The Catholic University of Korea. All subjects provided signed informed consent.

Electrophysiological measurements

Subjects were seated on a comfortable chair in a sound-attenuated room. A two-channel EEG device¹⁹ (Model: Amp AD8220; SOSO H&C, Kyungpook University, Daegu, Korea) was used to measure cortical activity in the frontal lobe for 5 min in a resting state with eyes closed. Dry electrodes were secured using a headband at Fp1 and Fp2 sites according to the 10–20 system. The reference electrode was placed on the right ear lobe. The sampling rate was 256 Hz. Online notch filtering (60 Hz) was

Table 1. Descriptive statistics in the study subjects

Variables	BF (<i>n</i> = 84)	HC (<i>n</i> = 25)	Statistics
	Mean (SD)		
Age (years)	44.95 (8.60)	43.44 (4.94)	$t = 0.84, P = 0.404$
Sex (<i>n</i> , male/female)	32/52	13/12	$\chi^2 = 1.54, \text{d.f.} = 1, P = 0.215$
Education (years)	12.52 (2.23)	15.36 (1.80)	$t = -5.81, P = 0.001$
Post-Traumatic Stress Disorder Checklist-5	47.13 (17.11)	—	—
Insomnia Severity Index	13.11 (6.66)	6.20(5.03)	$t = 4.79, P = 0.001$
Inventory of Complicated Grief (<i>n</i> = 73)	51.92 (14.91)	—	—
General Anxiety Disorder Checklist	10.63 (6.04)	2.28(3.60)	$t = 8.56, P = 0.001$
Spearman's rho-correlations in all subjects	EEG beta power	Education	Anxiety symptoms
EEG beta power	—	$P = 0.013$	$P = 0.025$
Education	—	—	$P = 0.002$
Anxiety symptoms	—	—	—

BF, bereaved families; EEG, electroencephalography; HC, healthy controls.

applied. Visual inspection was performed in artifact rejection before the preprocessing of the EEG data. Data were reanalyzed using MATLAB 2012 software (MathWorks, Inc., Natick, MA, USA.) using a Fast Fourier Transform with a bandpass filter of 1–50 Hz to calculate the amplitude values. The frequency power was calculated as the square of amplitudes: relative delta (1–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma (30–50 Hz) signals. The relative frequency power is a power ratio converted into the percentage of the whole frequency, which provides information about interactions between spectral oscillations in each band. Artifacts exceeding $\pm 100 \mu\text{V}$ were rejected at all electrode sites. Thirty artifact-free epochs (epoch length 2.048 s) were obtained for each subject. In order to identify possible EEG-signal artifacts caused by muscle tension, a correlation analysis was performed between the EEG power and the scores of restlessness, which was defined in accordance with the GAD Checklist as 'being so restless that it is hard to sit still.' No significant correlation was found.

Rating scales

PTSD Checklist-5

PTSD symptoms were assessed using the PTSD Checklist-5 (PCL-5), a 20-item self-report questionnaire, in which each item is answered on a 5-point (0–4) response scale.²⁰ The scores range from 0 to 80. Scores above 37 indicate severe symptoms. Internal consistency was defined by a Cronbach's alpha of 0.94. In the present study, the Cronbach's alpha

of the PCL-5 Korean version and its four sub-factors were as follows: re-experiencing, 0.85; avoidance, 0.76; negative alterations in cognition and mood, 0.88; hyperarousal, 0.87; all items, 0.94.

Inventory of Complicated Grief

Complicated grief was assessed using the Inventory of Complicated Grief (ICG), a 19-item self-report questionnaire, in which each item is answered on a 5-point (0–4) response scale.²¹ Total scores range from 0 to 76. A score above 25 indicates a high risk of symptoms. A strong internal consistency was defined by a Cronbach's alpha of 0.95.²² The internal consistency of the ICG Korean version was defined by a Cronbach's alpha of 0.87.²³

Insomnia Severity Index

Insomnia was assessed using the Insomnia Severity Index (ISI).²⁴ It consists of seven symptoms, rated on a scale from 0 to 4; the total scores range from 0 to 28, with higher scores indicating more severe forms of insomnia. A cut-off score of 15 has been used as the threshold for clinically significant insomnia, and a score below 8 has been used to define remission after treatment. The Cronbach's alpha of reliability coefficients of its three factors were around 0.80. The internal consistency of the ISI Korean version total score was defined by a Cronbach's alpha of 0.92.²⁵ Subgroups of the present study were classified according to symptom severity: normal sleep, 0–7; mild insomnia, 8–12; moderate insomnia, 13–17; and severe insomnia, 18–28.

GAD Checklist

Anxiety symptoms were assessed using the GAD Checklist. Seven symptoms are rated on a scale of 0–3 with a cut-off point of 10.²⁶ A strong internal consistency was defined by a Cronbach's alpha of 0.92. The validity of the GAD Korean version was defined by a Cronbach's alpha of 0.92.²⁷

Statistical analysis

We compared the demographic variables and the frontal EEG power of the BF and the HC using independent *t*-tests and the χ^2 -test. The analysis of Spearman's rho correlation was performed with bootstrapping (5000 times) among EEG beta power, education, and anxiety symptoms in all subjects. Years of education and anxiety symptoms were considered as covariates. Multivariate analysis of covariance was performed between the BF and the HC (frontal EEG power as a within-subject factor, BF vs HC as a between-subject factor), and using independent *t*-tests, we compared the EEG power of the low and high groups. The two BF subgroups were divided according to the cut-off values of psychological symptoms. We examined the between-subject effects (frontal EEG power as a dependent variable,

the four insomnia subgroups as a fixed factor) using multivariate analysis of variance. Multiple comparisons were corrected using the Bonferroni correction. In the BF, for between the psychological variables and EEG spectral power, Pearson's correlation analysis was performed with bootstrapping (5000 times). Statistical significance was set at $P < 0.05$ (two-tailed). The effect size calculation (Cohen's *d* and *F*-test) was performed by the G*POWER software 3.0.10 (Dusseldorf, Germany).

RESULTS

There were significant differences in the frontal beta power (Left: $f_{(1, 105)} = 7.77$, $P = 0.006$, Cohen's $d = 0.85$; Right: $f_{(1, 105)} = 9.80$, $P = 0.002$, Cohen's $d = 1.13$; Left + Right: $f_{(1, 105)} = 8.99$, $P = 0.003$, Cohen's $d = 0.99$) between the BF and the HC (Fig. 1a). Frontal beta power was higher in the BF than in the HC. Reduced delta activity (Left + Right: $f = 9.18$, $P = 0.003$, Cohen's $d = 0.99$) and increased alpha (Left + Right: $f = 6.20$, $P = 0.014$, Cohen's $d = 0.63$) was significant in the BF when compared to the HC (Fig. 1b). There were significant differences in beta and gamma activity between the subgroups of the BF that were divided according to

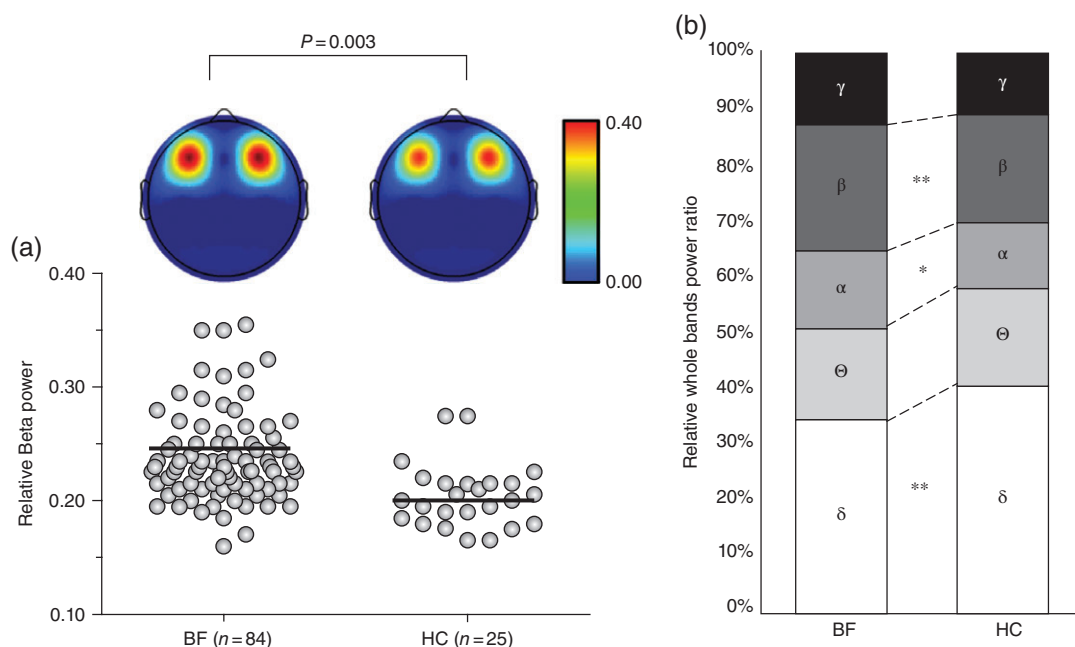


Figure 1. Comparisons of electroencephalography power between the bereaved families (BF) and healthy controls (HC). (a) Comparison of relative beta power. (b) Comparisons of whole bands relative power. Covariates are education and anxiety symptoms. (δ , delta; θ , theta; α , alpha; β , beta; γ , gamma).

their scores on psychological symptoms (Table 2). The gamma activity was significantly higher in the high-anxiety group ($t = 2.45$, $P = 0.016$, Cohen's $d = 4.00$) than in the low-anxiety group.

Within the BF, beta power was significantly lower in individuals with high insomnia than those with low insomnia ($t = 2.37$, $P = 0.020$, Cohen's $d = 4.00$) (Table 2). Moreover, the differences in beta activity were significant between the normal sleep and the severe insomnia group (Right: $f_{(3, 80)} = 3.32$, $P = 0.024$; Left + Right: $f_{(3, 80)} = 3.02$, $P = 0.034$) (effect size $F = 1.90$) (multiple comparison by Bonferroni correction, $P = 0.035$) (Fig. 2). Frontal beta activity decreased continuously with increasing symptom severity in the four subgroups of insomnia. Significant correlations in the BF were presented between PTSD symptom severity scores and insomnia severity scores ($r = 0.50$, $P = 0.001$), PTSD and complicated grief scores ($r = 0.75$, $P = 0.001$), and insomnia score and frontal beta power (Left: $r = -0.25$; $P = 0.024$; Right: $r = -0.26$, $P = 0.016$; Left + Right: $r = -0.26$, $P = 0.018$) (Fig. 3). In addition, significant correlations were presented between PTSD and anxiety symptom scores ($r = 0.63$, $P = 0.001$), complicated grief scores and anxiety symptom scores ($r = 0.59$, $P = 0.001$), and insomnia and anxiety symptom scores ($r = 0.39$, $P = 0.001$). However, the beta power was not correlated with PTSD symptom scores (Left: $r = -0.14$, $P = 0.222$; Right: $r = -0.16$, $P = 0.151$; Left + Right: $r = -0.15$, $P = 0.178$).

DISCUSSION

This study presents the first evidence of psychophysiological changes from the BF of the Sewol ferry disaster. The results of the present study revealed clinical and neurophysiological alterations in the BF of the Sewol ferry disaster. The first main finding was that the beta power was higher in the BF than in the HC. Second, beta power differed according to the severity of insomnia. More specifically, the frontal beta power of the BF demonstrated a significant negative correlation with their severity of insomnia.

The present study results showing high levels of PTSD symptoms, general anxiety, and beta power in EEG in traumatically bereaved subjects need to be interpreted with caution due to the differences in education levels and age ranges. It should be noted that our results were significant even after statistically controlling the level of education and anxiety symptoms. Our results replicated previous findings that identified higher beta power as a brain abnormality in panic and movement disorder.^{13,28}

The beta band could be associated with hyper-responsiveness owing to endogenous top-down processing of unexpected events. Increased beta activity in the resting state could indicate deficits of cognitive and behavioral flexibility, which could reflect the lack of inhibitory control in the frontal lobe of panic disorder patients.¹³ Cortical excitability following the enhanced beta band could result from impaired alertness and emotionally hyper-aroused states.²⁹ Begic *et al.*^{15,29} reported that beta activity

Table 2. Comparison of EEG power in the BF classified between group as low and high according to cut-off values of psychological symptoms scores

Classifications according to cut-off scores in BF	Low		High		EEG power comparisons (δ , θ , α , β , γ)
	<i>n</i> / Mean (SD) of symptom scores	Mean (SD) of EEG power	<i>n</i> / Mean (SD) of symptom scores	Mean (SD) of EEG power	
Post-Traumatic Stress Disorder Checklist-5	28 / 27.39 (8.11)	—	56 / 57.00 (10.58)	—	NS
Insomnia Severity Index	54 / 9.09 (4.08)	0.49 (0.01)	30 / 20.33 (3.52)	0.45 (0.01)	β / $P = 0.020$
Inventory of Complicated Grief ($n = 73$)	4 / 16.75 (4.92)	—	69 / 53.96 (12.54)	—	NS
General Anxiety Disorder Checklist	40 / 5.35 (2.74)	0.25 (0.01)	44 / 15.43 (3.74)	0.29 (0.01)	γ / $P = 0.016$

BF, bereaved families; EEG, electroencephalography; NS, not significant.

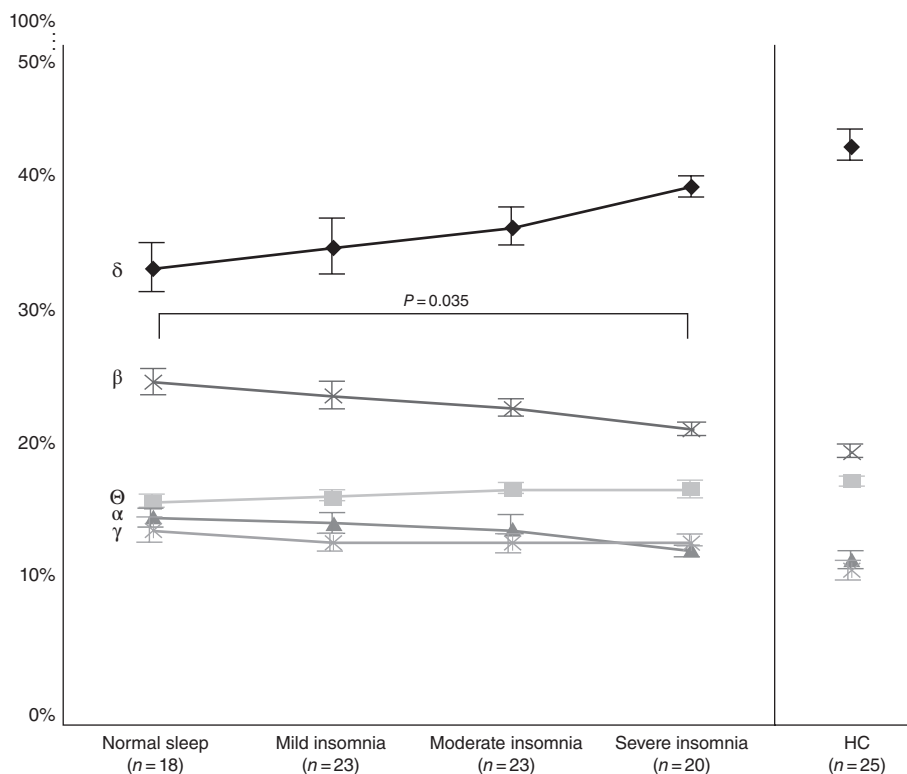


Figure 2. Proportional changes of electroencephalography relative power in the insomnia subgroup analysis. HC, healthy controls.

could be a reliable marker for neurophysiological changes, in which the enhanced beta activity by the cortical excitability in the frontal region was related to prolonged wakefulness.

Alternate explanations can be provided for the enhanced relative beta activity. One explanation is viewing it as a result of decreased delta activity. Perlis *et al.*³⁰ reported that the quantitative alteration of beta activity refers to changes of slow and spindle frequency activities, which suggest that decreased delta activity is linked to increased beta and sigma bands in subjects with insomnia. Furthermore, individuals with anxiety had demonstrated that increased alpha activity interplays with reduced delta activity.³¹ In the BF, individuals with high anxious symptoms showed an increased gamma activity, which, in anxiety, has been associated with negative emotions and worrying.³²

In the present study, beta power decreased gradually according to the severity of insomnia in BF. In addition, a significant reverse correlation was found between the severity of insomnia and frontal beta power. Previous sleep EEG studies have reported opposing results. In a study with PTSD patients, beta activity was higher during REM sleep.³³ In another

study, the beta activity in NREM sleep showed a positive correlation with the intensity and duration of combat exposure.³⁴ These previous results indicate that changes of beta frequency can reflect a state of hyperarousal during sleep. However, in an investigation that compared the EEG pattern between sleep-onset insomnia and sleep-maintenance insomnia, the beta (18–29.75 Hz) band was found to be significantly lower in the sleep-onset than in the sleep-maintenance.³⁵ This study suggests that sleep-onset insomnia may not be involved in the hyperarousal theory and that insomnia and its spectral characteristics ought to be observed in accordance with its subgroups.

However, our results from waking-state beta power are not comparable to the beta power that occurs during the sleep state. In waking state, a decreased beta power could represent an inattentive state of our participants.^{36–42} Insomnia has been associated with inattention.⁴³ In other respects, causes of change in the EEG band ratio have demonstrated that the altered beta power is related with the delta power changes in the BF. In our study, the correlation was significant between frontal beta power and insomnia symptom severity in the

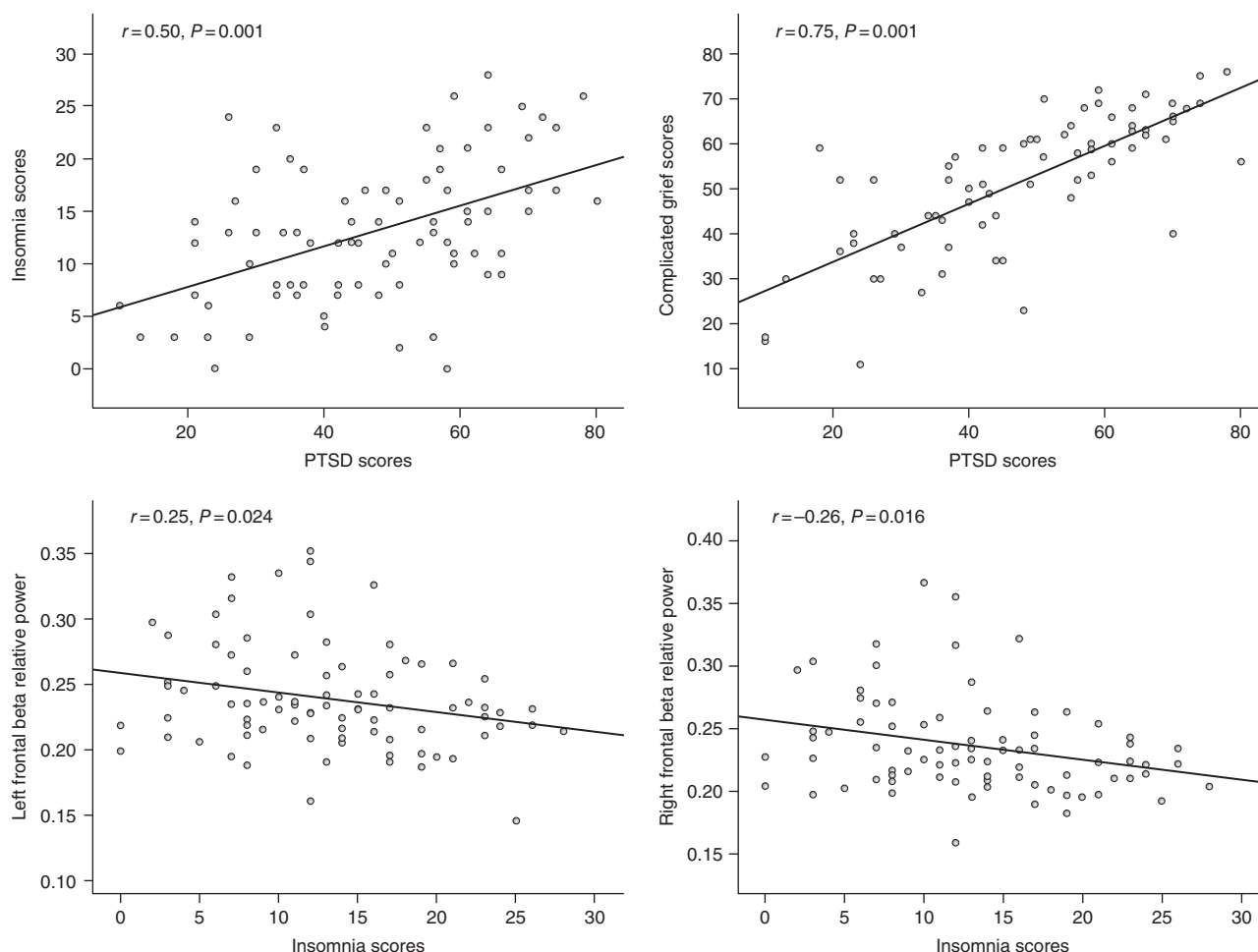


Figure 3. Pearson's correlations among post-traumatic stress disorder (PTSD) scores, insomnia scores, complicated grief scores, and frontal beta power in the bereaved families ($n = 84$) (Bootstrapping, $n = 5000$).

BF. Insomnia has been known to be pathognomonic in people with traumatic experiences.^{44,45} In the study by Maher *et al.*, 70–91% of patients with PTSD described their difficulty falling asleep,⁴⁶ some of whom also reported struggling from nightmares. The state of hypervigilance is usually observed through high-level beta activity in primary insomnia. However, contrary to our expectations, the present study found that beta power was reduced in the BF members with severe insomnia compared to those with mild insomnia. Lower beta activity is a reliable marker of attention deficit in depression.⁴⁷ In addition, attention-deficit/hyperactivity disorder is characterized by the reduction in beta activity.^{36,38,40,48} Specific insomnia cluster showed reduced spectral power of beta 1 and 2 for sustained

attention performance.⁴⁹ Inattentive domain in insomnia may be screened by reduction of beta activity. Our findings show that the decreased beta activity could be a potential marker of worsening insomnia in the BF of the Sewol ferry disaster's victims. However, the beta activity of the BF with severe insomnia does not differ from that of the HC. This contradictory finding might be an unprecedented phenomenon. The baseline levels in beta activity could have increased in the BF of the Sewol ferry disaster's victims in association with their insomnia symptoms. Complicated findings for insomnia previously reported^{50,51} that subjects who claimed sleep dysfunction as their primary symptom still suffered from insomnia even after other psychological symptoms had been alleviated.⁵² Persistent

symptoms of insomnia were associated with decreased volume of hippocampal subfield,⁵³ which is identical to the finding that non-PTSD patients with chronic insomnia showed a reduced hippocampal volume.⁵⁴ This could also be shown in the BF with severe symptoms of insomnia. EEG beta activity could be biomarker that reflects mental activity in various psychopathologies. Low beta activity, such as sensorimotor rhythm, has been known to enhance after neurofeedback training and result in the improvement of cognitive processing.⁵⁵ On the other hand, excessive beta activity (beta 1, 2, and 3) has been associated with intrusive thoughts between ruminators and non-ruminators.⁵⁶ These, therefore, can be considered as a mixed phase to negative and positive effect of neural response. In addition, although the beta activity was increased in the BF with normal sleep in comparison to HC or BF with severe insomnia, they showed several psychological problems, such as anxiety symptoms. It is difficult to posit that increased beta activity had a positive influence for the BF with normal sleep.

Three interpretations are provided for our results. First, increased EEG beta power in BF might reflect a coping strategy for extremely complex and stressful situations. Individuals with successful compensatory mechanism could have higher beta power and better quality of sleep. Second, sleep quality of BF could reflect a specific subtype of insomnia (i.e. primary insomnia with objective subtype). It was reported that primary insomnia patients with objective subtype had a lower degree of relative beta power than those with subjective subtype.⁵⁷ Third, if a higher shift of baseline levels in beta activity occurred in BF, a relative reduced beta power within BF could reflect inattention due to their insomnia symptoms. Several reports show that reduced beta power could reflect inattention.^{47,58,59}

The present study has limitations. We did not evaluate the handedness and more extensive symptoms, such as inattention and subtypes of insomnia, which could be reflected in electrophysiological measurements. EEG was not assessed during sleep. Assessments used in this study were based on self-reports rather than structured clinical interviews. In addition, subjects were not screened for psychiatric diagnoses. Further studies ought to recognize these limitations and work with a larger sample size. Future studies should also consider recruiting

control groups matched for sex, age range, anxiety symptoms, and education levels.

In summary, this study revealed that beta activity was higher in the BF than the HC group. High-levels of beta activity and its implications in neural mechanisms represent cortical excitability and hypervigilance in the BF. In addition, frontal beta power showed gradually decreasing patterns correlating with the severity of insomnia. Reduced beta activity specifies the worsening of insomnia symptoms in the BF of the Sewol ferry disaster's victims. Furthermore, the negative correlation between insomnia symptoms and frontal beta power probably suggest that frontal beta power is a potential marker of sleep disturbances. The symptom of insomnia is a crucial psychological problem in the BF of the Sewol ferry disaster's victims, which could be screened by EEG beta power.

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DISCLOSURE STATEMENT

The authors declare no conflicts of interest.

AUTHOR CONTRIBUTIONS

K.-I.J. contributed data acquisition, analysis, and completion of the manuscript. S.-H.L. and J.-H.C. designed the research and wrote the manuscript. M.S. performed data analysis. S.M.L., H.J.H., S.H., and J.-Y. J. collected data. S.-H.L. and J.-H.C. equally contributed as corresponding authors.

REFERENCES

1. Kang J-S, Ojha A, Lee M. Concentration monitoring with high accuracy but low cost EEG device. In: Arik S,

- Huang T, Lai W, Liu Q (eds). *Neural Information Processing. Lecture Notes in Computer Science*, Vol. 9492. Springer, Cham, 2015; 54–60.
2. Kim M-K, Kim S-P. Artifact removal from EEG signals using the total variation method. *2015 10th Asian Control Conference (ASCC)*. IEEE, Kota Kinabalu, 2015; 1–4.
 3. Suppiej A, Festa I, Bartolini L *et al*. Power spectral analysis of two-channel EEG in very premature infants undergoing heat loss prevention. *Neurophysiol. Clin.* 2014; 44: 239–244.
 4. Craig DA, Nguyen HT, Burchey HA. Two channel EEG thought pattern classifier. *Conf. Proc. IEEE Eng. Med. Biol. Soc.* 2006; 1: 1291–1294.
 5. Lobo I, Portugal LC, Figueira I *et al*. EEG correlates of the severity of posttraumatic stress symptoms: A systematic review of the dimensional PTSD literature. *J. Affect. Disord.* 2015; 183: 210–220.
 6. Todder D, Levine J, Abujumah A, Mater M, Cohen H, Kaplan Z. The quantitative electroencephalogram and the low-resolution electrical tomographic analysis in posttraumatic stress disorder. *Clin. EEG Neurosci.* 2012; 43: 48–53.
 7. Imperatori C, Farina B, Quintiliani MI *et al*. Aberrant EEG functional connectivity and EEG power spectra in resting state post-traumatic stress disorder: A sLORETA study. *Biol. Psychol.* 2014; 102: 10–17.
 8. Metzger LJ, Paige SR, Carson MA *et al*. PTSD arousal and depression symptoms associated with increased right-sided parietal EEG asymmetry. *J. Abnorm. Psychol.* 2004; 113: 324–329.
 9. Snyder SM, Hall JR. A meta-analysis of quantitative EEG power associated with attention-deficit hyperactivity disorder. *J. Clin. Neurophysiol.* 2006; 23: 440–455.
 10. Feshchenko VA, Veselis RA, Reinsel RA. Comparison of the EEG effects of midazolam, thiopental, and propofol: The role of underlying oscillatory systems. *Neuropsychobiology* 1997; 35: 211–220.
 11. van Lier H, Drinkenburg WH, van Eeten YJ, Coenen AM. Effects of diazepam and zolpidem on EEG beta frequencies are behavior-specific in rats. *Neuropharmacology* 2004; 47: 163–174.
 12. Lee S-H, Yoon S, Kim J-I, Jin S-H, Chung CK. Functional connectivity of resting state EEG and symptom severity in patients with post-traumatic stress disorder. *Prog. Neuropsychopharmacol. Biol. Psychiatry* 2014; 51: 51–57.
 13. Carvalho MR, Velasques BB, Cagy M *et al*. Electroencephalographic findings in panic disorder. *Trends Psychiatry Psychother.* 2013; 35: 238–251.
 14. Maes J, Verbraecken J, Willemsen M *et al*. Sleep misperception, EEG characteristics and autonomic nervous system activity in primary insomnia: A retrospective study on polysomnographic data. *Int. J. Psychophysiol.* 2014; 91: 163–171.
 15. Begic D, Hotujac L, Jokic-Begic N. Electroencephalographic comparison of veterans with combat-related post-traumatic stress disorder and healthy subjects. *Int. J. Psychophysiol.* 2001; 40: 167–172.
 16. van Wyk M, Thomas KGF, Solms M, Lipinska G. Prominence of hyperarousal symptoms explains variability of sleep disruption in posttraumatic stress disorder. *Psychol. Trauma* 2016; 8: 688–696.
 17. Fernandez-Mendoza J, Li Y, Vgontzas AN *et al*. Insomnia is associated with cortical hyperarousal as early as adolescence. *Sleep* 2016; 39: 1029–1036.
 18. Le Van Quyen M, Muller LE, Telenczuk B *et al*. High-frequency oscillations in human and monkey neocortex during the wake-sleep cycle. *Proc. Natl. Acad. Sci. U. S. A.* 2016; 113: 9363–9368.
 19. Lee S, Kruse J. Biopotential electrode sensors in ECG/EEG/EMG systems. *Analog Devices* 2008; 200: 1–2. [Cited 10 July 2017.] Available from URL: http://www.analog.com/media/en/technical-documentation/technical-articles/ecg-eeeg-emg_final.pdf.
 20. Blevins CA, Weathers FW, Davis MT, Witte TK, Domino JL. The posttraumatic stress disorder checklist for DSM-5 (PCL-5): Development and initial psychometric evaluation. *J. Trauma. Stress* 2015; 28: 489–498.
 21. Prigerson HG, Maciejewski PK, Reynolds CF *et al*. Inventory of complicated grief: A scale to measure maladaptive symptoms of loss. *Psychiatry Res.* 1995; 59: 65–79.
 22. Carmassi C, Shear MK, Massimetti G *et al*. Validation of the Italian version Inventory of Complicated Grief (ICG): A study comparing CG patients versus bipolar disorder, PTSD and healthy controls. *Compr. Psychiatry* 2014; 55: 1322–1329.
 23. Han DH, Lee JJ. Korean version of Inventory of Complicated Grief Scale: Psychometric properties in Korean adolescents. *J. Korean Med. Sci.* 2016; 31: 114–119.
 24. Chen P-Y, Yang C-M, Morin CM. Validating the cross-cultural factor structure and invariance property of the Insomnia Severity Index: Evidence based on ordinal EFA and CFA. *Sleep Med.* 2015; 16: 598–603.
 25. Cho YW, Song ML, Morin CM. Validation of a Korean version of the Insomnia Severity Index. *J. Clin. Neurol.* 2014; 10: 210–215.
 26. Spitzer RL, Kroenke K, Williams JBW, Löwe B. A brief measure for assessing generalized anxiety disorder: The GAD-7. *Arch. Intern. Med.* 2006; 166: 1092–1097.
 27. Seo JG, Cho YW, Lee SJ *et al*. Validation of the generalized anxiety disorder-7 in people with epilepsy: A MEPSY study. *Epilepsy Behav.* 2014; 35: 59–63.
 28. Engel AK, Fries P. Beta-band oscillations—Signalling the status quo? *Curr. Opin. Neurobiol.* 2010; 20: 156–165.
 29. Jokić-Begić N, Begić D. Quantitative electroencephalogram (qEEG) in combat veterans with post-traumatic stress disorder (PTSD). *Nord. J. Psychiatry* 2003; 57: 351–355.
 30. Perlis ML, Merica H, Smith MT, Giles DE. Beta EEG activity and insomnia. *Sleep Med. Rev.* 2001; 5: 363–374.
 31. Knyazev GG, Savostyanov AN, Levin EA. Alpha oscillations as a correlate of trait anxiety. *Int. J. Psychophysiol.* 2004; 53: 147–160.

32. Oathes DJ, Ray WJ, Yamasaki AS *et al.* Worry, generalized anxiety disorder, and emotion: Evidence from the EEG gamma band. *Biol. Psychol.* 2008; 79: 165–170.
33. Woodward SH, Murburg MM, Bliwise DL. PTSD-related hyperarousal assessed during sleep. *Physiol. Behav.* 2000; 70: 197–203.
34. Cohen DJ, Begley A, Alman JJ *et al.* Quantitative electroencephalography during rapid eye movement (REM) and non-REM sleep in combat-exposed veterans with and without post-traumatic stress disorder. *J. Sleep Res.* 2013; 22: 76–82.
35. Cervena K, Espa F, Perogamvros L, Perrig S, Merica H, Ibanez V. Spectral analysis of the sleep onset period in primary insomnia. *Clin. Neurophysiol.* 2014; 125: 979–987.
36. Satterfield JH, Schell AM, Backs RW, Hidaka KC. A cross-sectional and longitudinal study of age effects of electrophysiological measures in hyperactive and normal children. *Biol. Psychiatry* 1984; 19: 973–990.
37. Chabot RJ, Serfontein G. Quantitative electroencephalographic profiles of children with attention deficit disorder. *Biol. Psychiatry* 1996; 40: 951–963.
38. Clarke AR, Barry RJ, McCarthy R, Selikowitz M. EEG analysis in attention-deficit/hyperactivity disorder: A comparative study of two subtypes. *Psychiatry Res.* 1998; 81: 19–29.
39. Lazzaro I, Gordon E, Li W *et al.* Simultaneous EEG and EDA measures in adolescent attention deficit hyperactivity disorder. *Int. J. Psychophysiol.* 1999; 34: 123–134.
40. Mann CA, Lubar JF, Zimmerman AW, Miller CA, Muenchen RA. Quantitative analysis of EEG in boys with attention-deficit-hyperactivity disorder: Controlled study with clinical implications. *Pediatr. Neurol.* 1992; 8: 30–36.
41. Clarke AR, Barry RJ, McCarthy R, Selikowitz M. Hyperkinetic disorder in the ICD-10: EEG evidence for a definitional widening? *Eur. Child Adolesc. Psychiatry* 2003; 12: 92–99.
42. Chang A, Bosnyak DJ, Trainor LJ. Unpredicted pitch modulates beta oscillatory power during rhythmic entrainment to a tone sequence. *Front. Psychol.* 2016; 7: 327.
43. Weiss MD, Wasdell MB, Bomben MM, Rea KJ, Freeman RD. Sleep hygiene and melatonin treatment for children and adolescents with ADHD and initial insomnia. *J. Am. Acad. Child Adolesc. Psychiatry* 2006; 45: 512–519.
44. Harvey AG, Jones C, Schmidt DA. Sleep and posttraumatic stress disorder: A review. *Clin. Psychol. Rev.* 2003; 23: 377–407.
45. Nardo D, Högberg G, Jonsson C, Jacobsson H, Hällström T, Pagani M. Neurobiology of sleep disturbances in PTSD patients and traumatized controls: MRI and SPECT findings. *Front. Psychiatry* 2015; 6: 134.
46. Maher MJ, Rego SA, Asnis GM. Sleep disturbances in patients with post-traumatic stress disorder: Epidemiology, impact and approaches to management. *CNS Drugs* 2006; 20: 567–590.
47. Roh SC, Park EJ, Shim M, Lee SH. EEG beta and low gamma power correlates with inattention in patients with major depressive disorder. *J. Affect. Disord.* 2016; 204: 124–130.
48. Bresnahan SM, Anderson JW, Barry RJ. Age-related changes in quantitative EEG in attention-deficit/hyperactivity disorder. *Biol. Psychiatry* 1999; 46: 1690–1697.
49. Miller CB, Bartlett DJ, Mullins AE *et al.* Clusters of insomnia disorder: An exploratory cluster analysis of objective sleep parameters reveals differences in neurocognitive functioning, quantitative EEG, and heart rate variability. *Sleep* 2016; 39: 1993–2004.
50. Adelufosi A, Edet B, Arikpo D, Aquaisua E, Meremikwu MM. Cognitive behavioral therapy for post-traumatic stress disorder, depression, or anxiety disorders in women and girls living with female genital mutilation: A systematic review. *Int. J. Gynaecol. Obstet.* 2017; 136 (Suppl. 1): 56–59.
51. Jansson-Frojmark M, Norell-Clarke A. Cognitive behavioural therapy for insomnia in psychiatric disorders. *Curr. Sleep Med. Rep.* 2016; 2: 233–240.
52. Zayfert C, DeViva JC. Residual insomnia following cognitive behavioral therapy for PTSD. *J. Trauma. Stress* 2004; 17: 69–73.
53. Neylan TC, Mueller SG, Wang Z *et al.* Insomnia severity is associated with a decreased volume of the CA3/dentate gyrus hippocampal subfield. *Biol. Psychiatry* 2010; 68: 494–496.
54. Riemann D, Voderholzer U, Spiegelhalder K *et al.* Chronic insomnia and MRI-measured hippocampal volumes: A pilot study. *Sleep* 2007; 30: 955.
55. Kober SE, Witte M, Stangl M, Valjamae A, Neuper C, Wood G. Shutting down sensorimotor interference unblocks the networks for stimulus processing: An SMR neurofeedback training study. *Clin. Neurophysiol.* 2015; 126: 82–95.
56. Ferdek MA, van Rijn CM, Wyczesany M. Depressive rumination and the emotional control circuit: An EEG localization and effective connectivity study. *Cogn. Affect. Behav. Neurosci.* 2016; 16: 1099–1113.
57. Krystal AD, Edinger JD, Wohlgemuth WK, Marsh GR. NREM sleep EEG frequency spectral correlates of sleep complaints in primary insomnia subtypes. *Sleep* 2002; 25: 630–640.
58. Roh SC, Park EJ, Park YC *et al.* Quantitative electroencephalography reflects inattention, visual error responses, and reaction times in male patients with attention deficit hyperactivity disorder. *Clin. Psychopharmacol. Neurosci.* 2015; 13: 180–187.
59. Dupuy FE, Clarke AR, Barry RJ, McCarthy R, Selikowitz M. Girls with attention-deficit/hyperactivity disorder: EEG differences between DSM-IV types. *Clin. EEG Neurosci.* 2011; 42: 1–5.